Orthogonal Terrain-Following Coordinates for the Higher-Order Parabolic Equation: Wide-Angle Propagation Over Irregular Terrain With Discrete Transparent Boundaries

Mikhail S. Lytaev

***Abstract*—Parabolic equation (PE) solvers usually represent terrain by a staircase or by the Beilis–Tappert shift map. We map the propagation domain onto a rectangle by orthogonal terrain-following coordinates built from the tapered harmonic extension of the terrain. In these coordinates rational (Padé) propagators of arbitrary order apply without additional angle restrictions, and the discrete nonlocal transparent boundary condition (NLBC) remains exact, unlike with the shift map or conformal maps. The terrain enters the vertical Numerov scheme as a variable coefficient and a weight. A flux-conserving range step removes the amplitude drift caused by the range-dependent metric. A compact closure of the impedance condition has a reflection error bounded uniformly in the ground parameters. A discrete dispersion analysis proves local equivalence with the flat-terrain scheme. Applied to shift-map PEs, it reveals a slope error that does not vanish under grid refinement, even with an exact square root. On flat ground, these PEs reach 6–35° at an accuracy of 10^{-4} rad per wavelength, whereas the proposed scheme reaches $82°$, is 20 to 560 times cheaper, and has no maximum angle. Against exact solutions for beams reflected by conducting and impedance slopes at up to $70°$, its error is at most $3 \cdot 10^{-3}$, three orders of magnitude below that of the staircase and of the shift maps. Absorbing layers must be tuned to the field's angular content and still leave errors up to three orders of magnitude above those of the NLBC. A real-terrain example with weather model refractivity is presented.**

***Index Terms*—Parabolic equation, irregular terrain, terrain-following coordinates, Padé approximation, nonlocal boundary conditions, impedance boundary condition, tropospheric propagation, numerical dispersion.**

I. Introduction

THE parabolic equation (PE) is the standard tool for predicting radio-wave propagation over irregular terrain in an inhomogeneous troposphere [1]. Finite-difference higher-order PE methods approximate the one-way propagator by a rational (Padé) function of the vertical operator [2]. Combined with a fourth-order vertical discretization [3], an automated choice of the order and of the grid [4], [5], and discrete nonlocal transparent boundary conditions (NLBCs) [6], [7], they resolve wide propagation angles on coarse grids and truncate the domain just above the region of interest.

Terrain, however, is usually still represented by a staircase: the field is set to zero below the terrain column by column. A staircase converges only at first order, scatters spuriously at every step, and for steep slopes demands a vertical step much smaller than the propagation angle requires. As shown below, the staircase error at wide angles does not decrease appreciably even with very fine grids.

The alternative is a change of variables that makes the terrain a coordinate line. The Beilis–Tappert shift map [8], combined with a phase factor, maps the narrow-angle (NA) PE over terrain exactly onto the PE over flat ground with an extra potential. It underlies TPEM, TEMPER and APM [9]–[11], as well as the acoustic PERUSE model [12] and its successors [13]–[15]. For the Helmholtz equation, however, the shear produces a mixed derivative, which the phase factor removes only at the paraxial level.

Recent shift-map finite-difference PEs: Recent work on finite-difference (FD) shift-map PEs falls into a few groups.

- *Wide-angle (WA) square roots with Crank–Nicolson (CN) marching.* These replace the square root by the Claerbout $[1/1]$ approximation, by the Greene approximation or by its Padé $[2/2]$ approximation, and march in range with CN [16]–[22]. Discretizations use three- or five-point differences in height and grids of $\Delta x = \Delta z = \lambda/2$ or $\Delta x \approx 5\lambda$, $\Delta z = \lambda/10$.
- *Reported limits.* On flat ground the reported angles are about $15°$ (NA), 36–$40°$ (Claerbout), $47°$ (Greene) and $55°$ (Padé $[2/2]$) [18], [19], [23]. Over terrain, the WA shift map is limited to slopes of 10–20° [16], [18], [21].
- *Wide-angle versions can be worse than NA.* The WA shift map is reported to be less accurate than the NA one on lee-side slopes steeper than about $5°$ [24], [25]. Gilbert [15] traced this to the vertical wavenumber in sheared coordinates being unphysical.
- *Required aperture.* The maximum angle required over terrain is the flat-ground aperture plus twice the slope [26].
- *Domain truncation.* All these works truncate the domain from above with absorbing layers: Tukey or Hanning windows over the upper 30% of the domain [20], [27], a complex refractive index over the upper third [14], or perfectly matched layers [28], [29].
- *Accuracy analyses.* Grid-dispersion analyses of FD PEs [30] quantify the discretization error, but not the grid-independent error of the shift map.

Other approaches: The rotated PE [31], [32] is exact for piecewise-constant slopes but needs field interpolation at every

M. S. Lytaev is with [affiliation to be completed] (e-mail: mikelytaev@gmail.com).

slope break. Conformal maps preserve the Laplacian and have been combined with narrow-angle split-step Fourier solvers [12], [33], with bent optical waveguides [34], and with wide-angle rational PEs for water waves [35]. Local orthogonal transformations with exact one-way methods were developed for bounded acoustic waveguides [36]. Kampanis [37] used a terrain-fitting map that is the identity near the top so that an NLBC applies, but with a narrow-angle finite-element PE. To our knowledge, no method combines an exact (Helmholtz-level) treatment of the terrain, rational approximations of arbitrary order, and an exact discrete NLBC.

This paper provides such a method. Its contributions are as follows.

1) An explanation of why orthogonal coordinates are the natural setting for an arbitrary-order rational propagator with the NLBC (Section III): the shift map and conformal maps are not the identity near the upper boundary, whereas orthogonal coordinates can be, and keep the structure of the flat problem.
2) A construction of orthogonal terrain-following coordinates that coincide with Cartesian coordinates above a transition layer (Section IV).
3) The transformed one-way equation (Section V). Each factor of the rational propagator reduces to the Numerov scheme with a variable coefficient and one additional weight. A flux-conserving range step, written in a form that is the identity in the exterior, accounts for the range dependence of the metric.
4) A compact fourth-order closure of the Leontovich impedance condition on the terrain (Section VI-A). Its discrete reflection error is bounded uniformly in the ground parameters.
5) A discrete dispersion analysis (Section VII) proving local equivalence with the flat-terrain scheme and bounding the reflection error of the ground, and a proof that the discrete NLBC is unchanged (Section VIII).
6) A quantitative comparison with the NA and WA shift-map FD PEs (Section IX). The comparison uses the same discrete dispersion analysis: the grid-independent error floor of the shift map, the maximum angle reachable at a given accuracy, and the computational cost for a given angle. It is confirmed against exact solutions.
7) A comparison of the discrete NLBC with the absorbing layers used in the literature (Section X).
8) Validation against exact wide-angle solutions over conducting and impedance slopes, and a demonstration over real terrain (Section XI).

II. Problem Statement

Consider the two-dimensional Helmholtz equation for a time-harmonic field $\psi(x,z)$ ($e^{-i\omega t}$ dependence)

$$\nabla^2\psi + k^2(x,z)\psi = 0 \tag{1}$$

in the domain $\Omega = \{z > B(x)\}$ above the terrain $B(x)$, where z is the height, $k = k_0 n$, and n is the modified refractive index, including the Earth flattening. The ground is either perfectly conducting, with $\psi = 0$ on $z = B(x)$ for horizontal (H) and $\partial_n\psi = 0$ for vertical (V) polarization, or it satisfies the Leontovich impedance condition [1]

$$\partial_n\psi + \alpha\psi = 0, \quad \alpha = ik_0\Delta_g, \quad \Delta_g = \begin{cases} \sqrt{\varepsilon - 1}, & \text{H}, \\ \sqrt{\varepsilon - 1}/\varepsilon, & \text{V}, \end{cases} \tag{2}$$

where ε is the complex relative permittivity of the ground and n is the normal directed into Ω. A plane wave incident at the grazing angle ϕ to the local ground is reflected with $R(\phi) = (\sin\phi - \Delta_g)/(\sin\phi + \Delta_g)$. Above some height the medium does not depend on x, and the domain is truncated there by an NLBC.

On flat ground let $L = \beta^{-2}\left(\partial_z^2 + k^2\right) - 1$ with a reference wavenumber β, and let $u = \psi e^{-i\beta x}$. The outgoing field obeys $u_x = i\beta(\sqrt{1+L} - 1)u$, and one range step is

$$e^{i\beta\Delta x\left(\sqrt{1+L}-1\right)} \approx R(L) = \prod_{l=1}^{p} \frac{1 + a_l L}{1 + b_l L}. \tag{3}$$

Each factor is a tridiagonal solve

$$\begin{aligned} b\,\delta^2 v + (1 + \alpha\delta^2)\left[\tau^2(1 + b\nu)v\right] \\ = a\,\delta^2 u + (1 + \alpha\delta^2)\left[\tau^2(1 + a\nu)u\right], \end{aligned} \tag{4}$$

where $\tau = \beta\Delta z$, $\alpha = 1/12$, $\nu = k^2/\beta^2 - 1$, and $\delta^2 u_j = u_{j+1} - 2u_j + u_{j-1}$. This is the Numerov–Padé scheme of [3]. The discrete NLBC [6] expresses the uppermost unknown of every factor through the history of the boundary values of all factors,

$$u^n_{N-1,l} - \sum_{l' \le l} Q^{(0)}_{ll'} u^n_{N,l'} = \sum_{m=1}^{n} \sum_{l'} Q^{(m)}_{ll'} u^{n-m}_{N,l'}. \tag{5}$$

The kernels $Q^{(m)}$ are the inverse z-transform of a matrix function of a_l, b_l and of the discrete exterior equation. They depend only on the exterior scheme and on the chain of factors.

III. Why Orthogonal Coordinates

A. Existing Transformations and the NLBC

The discrete NLBC (5) is derived for an exterior in which the discrete equation does not depend on range. A transformation that flattens the terrain must therefore become the identity at some height below the upper boundary. The existing transformations do not.

- *Shift map.* The substitution $\zeta = z - T(x)$ moves the whole column, including the upper boundary, with the terrain. The transformed index $n^2(\zeta + T) - 2\zeta T''$ and the phase factor $e^{ik_0 T'\zeta}$ depend on range at every height and grow with ζ. This is why shift-map solvers truncate the domain with absorbing layers (Section X).
- *Conformal maps.* An analytic function that equals $w + c$ on an open set equals $w + c$ everywhere. A conformal map that flattens the terrain is therefore nowhere the identity, and the exterior index $n^2|F'|^2$ depends on range at every height. For a feature of horizontal scale ℓ, $|F'|^2 - 1$ decays like $e^{-2\pi\eta/\ell}$, so a $10\,\mathrm{km}$ hill still leaves a residual of order 10^{-1} at a height of $1\,\mathrm{km}$.

The staircase is compatible with the NLBC, but it does not treat the terrain as a boundary. We therefore look for a map

that is the identity above a transition layer and, below it, keeps the structure of the flat problem that the rational propagator relies on. Orthogonal coordinates provide both.

B. Orthogonal Curvilinear Coordinates

Let $(x, z) = (X(\xi,\eta), Z(\xi,\eta))$ map the computational rectangle onto the propagation domain, with the terrain at $\eta = 0$. Its tangent vectors are $\mathbf{a}_\xi = (X_\xi, Z_\xi)$, along the lines $\eta = \text{const}$, and $\mathbf{a}_\eta = (X_\eta, Z_\eta)$, along the lines $\xi = \text{const}$. The coordinates are orthogonal if $\mathbf{a}_\xi \cdot \mathbf{a}_\eta = 0$ everywhere. The arc length then has no cross term,

$$ds^2 = h_\xi^2\, d\xi^2 + h_\eta^2\, d\eta^2, \qquad h_\xi = |\mathbf{a}_\xi|, \quad h_\eta = |\mathbf{a}_\eta|. \tag{6}$$

The scale factors have a direct geometric meaning: a computational cell $\Delta\xi \times \Delta\eta$ is, to leading order, a rectangle with sides $h_\xi\Delta\xi$ along the level curve and $h_\eta\Delta\eta$ across it, rotated by the local slope of the level curve. Their ratio $r = h_\eta/h_\xi$ is the anisotropy of the cell; $r = 1$ for a conformal map.

Proposition 1 (Orthogonal coordinates): In orthogonal coordinates (1) reads

$$\partial_\xi\left(r\partial_\xi\psi\right) + \partial_\eta\left(\frac{1}{r}\partial_\eta\psi\right) + h_\xi h_\eta k^2\psi = 0, \tag{7}$$

and the derivative along the normal to a level curve is $\partial_n = h_\eta^{-1}\partial_\eta$.

Proof: The gradient is $\nabla\psi = h_\xi^{-2}\mathbf{a}_\xi\psi_\xi + h_\eta^{-2}\mathbf{a}_\eta\psi_\eta$, the area element is $h_\xi h_\eta\, d\xi\, d\eta$, and the divergence theorem gives $\nabla^2\psi = (h_\xi h_\eta)^{-1}\left[\partial_\xi(h_\eta h_\xi^{-1}\partial_\xi\psi) + \partial_\eta(h_\xi h_\eta^{-1}\partial_\eta\psi)\right]$. Multiply by $h_\xi h_\eta$. The unit normal to $\eta = \text{const}$ is $\mathbf{a}_\eta/h_\eta$. ∎

Equation (7) has the three properties the method needs.

- *No mixed derivative.* In general coordinates the metric has an off-diagonal term, which produces $\partial_\xi\partial_\eta\psi$. For the shift map, for instance, $\psi_{xx} - 2s\psi_{x\zeta} + (1+s^2)\psi_{\zeta\zeta} + k^2\psi = 0$ with $s = T'$. With a mixed derivative the incident and reflected vertical wavenumbers of a plane wave at the boundary are no longer opposite, and no function of a single vertical operator is the exact propagator. In (7) the range and height derivatives are separated. After the Liouville scaling of Section V, the equation takes the form $\varphi_{\xi\xi} + B\varphi = 0$ with a vertical operator B, and the exact step is $\exp(i\Delta\xi\sqrt{B})$, which is approximated by (3) exactly as on flat ground.
- *Local boundary conditions.* Since $\partial_n = h_\eta^{-1}\partial_\eta$ on $\eta = 0$, the Dirichlet, Neumann and impedance conditions remain conditions in η alone at a single node, and the vertical operator is symmetric with them. For frozen coefficients, $\eta \mapsto -\eta$ is a symmetry of (7), so the Dirichlet (Neumann) half-space solutions are odd (even) images, as on flat ground.
- *Local plane waves.* For frozen scale factors, $e^{i(\kappa_\xi\xi + \kappa_\eta\eta)}$ solves (7) if $(\kappa_\xi/h_\xi)^2 + (\kappa_\eta/h_\eta)^2 = k^2$, i.e. $\kappa_\xi = h_\xi k\cos\theta$ and $\kappa_\eta = h_\eta k\sin\theta$, where θ is the physical angle between the wave and the level curve. Locally the transformed problem is the flat problem with wavenumber $h_\xi k$ and vertical stretch r, and the grid $(\Delta\xi, \Delta\eta)$ acts as the physical grid $(h_\xi\Delta\xi, h_\eta\Delta\eta)$ aligned with the terrain. This is the basis of the dispersion analysis of Section VII.

Orthogonality leaves one family of coordinate lines free: once the level curves $\eta = \text{const}$ are chosen, the lines $\xi = \text{const}$ are their orthogonal trajectories and are determined uniquely. We use this freedom to make the level curves coincide with the terrain at $\eta = 0$ and with horizontal lines above the transition layer (Section IV). The price, compared with a conformal map, is the anisotropy $r \neq 1$, which enters the scheme only as a variable coefficient.

IV. Orthogonal Terrain-Following Coordinates

A. Level Curves

The coordinate lines $\eta = \text{const}$ are the curves $z = F(x,\eta) = \eta + z_{\text{ref}} + u(x,\eta)$, where u is the harmonic extension of $b = B - z_{\text{ref}}$, tapered to zero:

$$\hat{u}(\kappa,\eta) = \hat{b}(\kappa)\, w(\eta/L)\, e^{-|\kappa|\eta}, \tag{8}$$

$$w(t) = \begin{cases} 1 - t^3(10 - 15t + 6t^2), & 0 \le t \le 1, \\ 0, & t > 1. \end{cases}$$

Here $\hat{}$ denotes the Fourier transform in x, and L is the height of the transition layer. The construction has four properties.

- *Exact boundary.* $F(x,0) = B(x)$.
- *Identity above the layer.* $F = \eta + z_{\text{ref}}$ for $\eta \ge \eta_\ell = L$: the coordinates are Cartesian, up to a constant shift, above the transition layer.
- *Near-conformality near the ground.* Near the ground u is harmonic, so a feature of horizontal scale ℓ decays over a height of about ℓ. Small-scale relief stays near the ground, as with a conformal map, and only the large-scale relief reaches the taper.
- *Regularity.* With $u = wU$, the map is regular when $F_\eta = 1 + wU_\eta + w'U/L > 0$. For $z_{\text{ref}} = \max B$ we have $U \le 0$ (maximum principle) and $w' \le 0$, so the taper term is non-negative for every L, and $F_\eta \ge 1 - \max|U_\eta|$ is a bound on the smoothed slopes. For $z_{\text{ref}} = \overline{B}$ the bound acquires the term $-\frac{15}{8}\max(B - z_{\text{ref}})/L$.

The terrain samples are filtered with a Gaussian of width σ to remove the corners of the piecewise-linear profile, and are extended evenly at $x = 0$. The even extension makes $u_x(0,\eta) = 0$, so the starting line $\xi = 0$ is the vertical line $x = 0$ and the antenna pattern is prescribed in height as usual.

B. Orthogonal Trajectories and Metrics

The level curves fix η. It remains to find the lines $\xi = \text{const}$ and the scale factors. A node (ξ,η) is mapped to

$$x = X(\xi,\eta), \qquad z = F\big(X(\xi,\eta),\eta\big),$$

where X is the abscissa of the line $\xi = \text{const}$ at the level η; the second relation says that the node lies on the level curve η, so that $Z(\xi,\eta) = F(X(\xi,\eta),\eta)$ in the notation of Section III. Since F is known from (8), the only unknown is X. The tangents to the two coordinate lines are

$$\mathbf{a}_\xi = (X_\xi, Z_\xi) = P\,(1, F_x),$$
$$\mathbf{a}_\eta = (X_\eta, Z_\eta) = (X_\eta,\, F_x X_\eta + F_\eta),$$

by the chain rule, where $P = X_\xi$ and F_x, F_η are taken at $x = X$.

Trajectories. Setting the scalar product of the two tangents to zero gives an ordinary differential equation for each line $\xi = \text{const}$:

$$X_\eta = G(X,\eta) \equiv -\frac{F_x F_\eta}{1+F_x^2}, \qquad X(\xi,\eta_\ell) = \xi. \tag{9}$$

Above the layer $G = 0$ and the lines are vertical, so they are labeled by their abscissa there. Integrating (9) downward from η_ℓ to 0 traces each line through the layer, crossing every level curve at a right angle. The trajectories are integral curves of the non-vanishing field $\nabla\eta$ and never cross.

Scale factors. The scale factors are the lengths of the two tangents. With $X_\eta = G$ we have $F_x X_\eta + F_\eta = F_\eta/(1+F_x^2)$, and

$$h_\eta = \frac{F_\eta}{\sqrt{1+F_x^2}}, \quad h_\xi = P\sqrt{1+F_x^2}, \quad P_\eta = G_x(X,\eta)P, \tag{10}$$

with $P(\xi,\eta_\ell) = 1$. The equation for P follows by differentiating (9) in ξ. Both factors have a simple meaning. Neighboring level curves are $F_\eta\, d\eta$ apart vertically; projected on the normal, which is tilted by $\arctan F_x$, this gives h_η. It depends on F at the node only. Neighboring trajectories are $P\, d\xi$ apart horizontally; converted to arc length along the level curve, this gives h_ξ. Through P, which accumulates the convergence ($P<1$) or divergence ($P>1$) of the trajectories on their way down, h_ξ depends on the whole path above the node. As a check, $h_\xi h_\eta = PF_\eta$ equals the Jacobian of the map, and the anisotropy is $r = F_\eta/[P(1+F_x^2)]$.

Computation. The function $\hat{b}$ is computed once by FFT. On each level, one inverse FFT of $\hat{u}$ and of its products with $i\kappa$, $-\kappa^2$ and ∂_η gives F_x, F_η, F_{xx} and $F_{x\eta}$ on a fine uniform grid in x; they are interpolated to the current X to form G and G_x. Equations (9)–(10) are integrated downward with the fourth-order Runge–Kutta method for all columns at once, on coarse levels, and the metrics are then interpolated to the computational nodes with cubic splines. The map takes seconds to build. Orthogonality of the computed grid was verified to $2\cdot 10^{-4}$, the level of the finite-difference check itself. Fig. 1 shows the resulting grids.

V. Transformed One-Way Equation

A. Liouville Transformation

The range dependence of r enters (7) through $\partial_\xi(r\partial_\xi)$. Setting $\psi = r^{-1/2}\varphi$ and using $\partial_\xi(r\partial_\xi(r^{-1/2}\varphi)) = r^{1/2}\varphi_{\xi\xi} - (r^{1/2})_{\xi\xi}\varphi$, we obtain

$$\begin{aligned} &\varphi_{\xi\xi} + B\varphi = 0, \\ &B\varphi = r^{-1/2}\partial_\eta\left(\frac{1}{r}\partial_\eta\left(r^{-1/2}\varphi\right)\right) + \left(k^2 h_\xi^2 - V\right)\varphi, \end{aligned} \tag{11}$$

with $V = (r^{1/2})_{\xi\xi}/r^{1/2}$, a term of relative size $(\lambda/\ell)^2$. The operator B is symmetric in $L^2(d\eta)$. Dirichlet and Neumann conditions are invariant. Since $\partial_n = h_\eta^{-1}\partial_\eta$ on $\eta = 0$, the impedance condition (2) becomes the Robin condition

$$\partial_\eta\psi + \gamma\psi = 0, \qquad \gamma(\xi) = \alpha\, h_\eta(\xi,0), \tag{12}$$

with a coefficient that varies along the ground. The scheme below is written for ψ, so (12) is imposed on ψ directly.

B. Flux-Conserving Range Step

The one-way factorization of (11) retains the transport term, $\varphi_\xi = i\sqrt{B}\varphi - \frac{1}{4}B^{-1}B_\xi\varphi + \ldots$, which conserves the energy flux $\|B^{1/4}\varphi\|^2$ rather than $\|\varphi\|^2$. In the usual PE for a slowly varying medium this term is neglected. Here it is not negligible: the coordinate lines rotate with height, so a beam changes its angle to the local marching direction. The plain step then loses amplitude, by 2–6% over a few hundred meters for beams reflected at 40–50° in the tests of Section XI-A, in agreement with the WKB factor $(\cos\theta_1/\cos\theta_2)^{1/2}$. We use the flux-conserving step [38], [39]. Since the middle factor commutes with $B_{n+1/2}^{\pm 1/4}$, it can be written as

$$\begin{aligned} \varphi_{n+1} &= \left[B_{n+1}^{-1/4}B_{n+1/2}^{1/4}\right] e^{i\Delta\xi\sqrt{B_{n+1/2}}} \left[B_{n+1/2}^{-1/4}B_n^{1/4}\right]\varphi_n \\ &= B_{n+1}^{-1/4} e^{i\Delta\xi\sqrt{B_{n+1/2}}} B_n^{1/4}\varphi_n. \end{aligned} \tag{13}$$

Each bracket is the identity wherever its two columns coincide, in particular in the whole exterior. The middle factor is approximated by (3) with $L = B/\beta^2 - 1$.

The factors $(1+L)^{\pm 1/4}$ are approximated by rational functions. Their Padé approximants have poles and zeros on the branch cut $L < -1$, i.e., inside the evanescent spectrum. There a pair of factors evaluated at slightly different columns can amplify a spurious mode; we observed a 14% localized error after 600 m. Following [40], we take the $[q/q]$ Padé approximant of $(1+Y)^{1/4}$ with $Y = (1+L)e^{-i\gamma} - 1$ and $\gamma = 45°$. Rewritten as factors $(1+cL)/(1+dL)$, its poles and zeros lie off the real axis. The constant prefactor cancels within each pair, and only the logarithmic derivative matters, which for $q = 2$ is accurate to 5% up to 45°. Since the correction itself is a few percent, the resulting error is of order 10^{-3}.

C. Factors in Terms of the Physical Field

Lemma 1: Let $\varphi = r^{1/2}\psi$, $m = r^2$ and $\nu = (k^2h_\xi^2 - V)/\beta^2 - 1$. Then $r^{3/2}\beta^2(1+cL)\varphi = c\, r\,\partial_\eta(r^{-1}\partial_\eta\psi) + \beta^2 m(1+c\nu)\psi$.

Proof: Insert (11) into $\beta^2(1-c)\varphi + cB\varphi$ and multiply by $r^{3/2}$. ∎

Every factor of (13) is therefore the flat scheme (4), with ∂_η^2 replaced by the operator $r\partial_\eta r^{-1}\partial_\eta$ with the variable coefficient $r = h_\eta/h_\xi$, and with an additional weight m. Between columns the field is rescaled by $(m_{\text{old}}/m_{\text{new}})^{1/4}$.

VI. Discretization

On the uniform grid $\eta_j = j\Delta\eta$ each factor reads

$$\begin{aligned} &c\,\delta_r^2 v + (1+\alpha\delta_r^2)\left[\tau^2 m(1+c\nu)v\right] \\ &\qquad = a\,\delta_r^2 u + (1+\alpha\delta_r^2)\left[\tau^2 m(1+a\nu)u\right] \end{aligned} \tag{14}$$

with $\tau = \beta\Delta\eta$ and $\delta_r^2 u_j = r_j\left[(u_{j+1}-u_j)/r_{j+1/2} - (u_j - u_{j-1})/r_{j-1/2}\right]$. For $g = \ln r$ and $D = r\partial r^{-1}\partial$, a Taylor expansion gives $\delta_r^2/\Delta\eta^2 = D + \frac{\Delta\eta^2}{12}D^2 + \Delta\eta^2 E + O(\Delta\eta^4)$, where $E\psi = \frac{1}{24}\left[(g'^2+g'')\psi'' + (g''' + g'g'' - g'^3)\psi'\right]$. The Numerov weight cancels the D^2 term. The scheme is thus fourth-order in the wave part, with a second-order error of relative size $(\Delta\eta/\ell_\eta)^2/24$, negligible since $\ell_\eta \gtrsim L$.

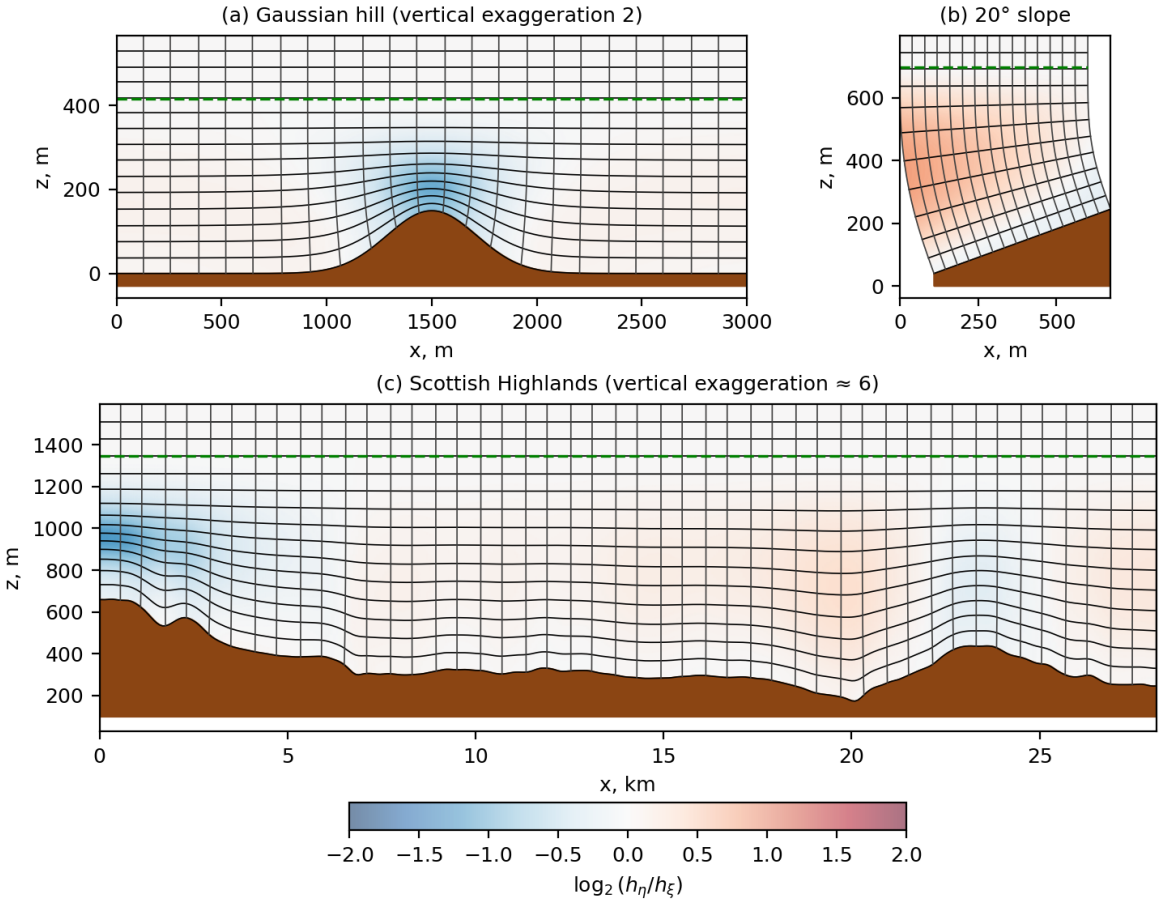


Fig. 1. Orthogonal terrain-following coordinates. Black: level curves $\eta = \text{const}$; grey: trajectories $\xi = \text{const}$; color: $\log_2(h_\eta/h_\xi)$; dashed: top of the transition layer, above which the coordinates are Cartesian. (a) Gaussian hill, $z_{\text{ref}} = \overline{B}$, $L = 400\,\text{m}$. (b) 20° slope without the even extension at $x = 0$; the starting trajectory is then curved. (c) Scottish Highlands profile of Section XI-D. Panels (a) and (c) are vertically exaggerated, so the right angles appear distorted.

Algorithm 1: one range step

1: interpolate (ν, m, r) at ξ_n, $\xi_{n+1/2}$, ξ_{n+1}
2: $\psi \leftarrow C_1\psi$ (first bracket of (13); rows below η_ℓ plus a margin)
3: **for** $l = 1, \ldots, p$ **do**
4: solve (14) with (a_l, b_l) at $\xi_{n+1/2}$ and the NLBC row (5)
5: **end for**
6: $\psi \leftarrow C_2\psi$ (second bracket of (13))

At the ground the Dirichlet unknown is eliminated from the system rather than imposed by a unit row; a unit row next to entries of order $\alpha\tau^2$ is satisfied by a pivoting solver only up to amplified round-off. The Neumann condition uses a mirror node. The impedance closure is described in Section VI-A. The coefficients ν, m, r are tabulated on coarse columns, only below the Cartesian region, and are interpolated linearly in range; the main chain is evaluated at the midpoint. One step (Algorithm 1) costs $p + 4q$ tridiagonal solves, and the $4q$ flux solves are restricted to the transition layer.

A. Impedance Ground

For the Robin condition (12) a ghost node u_{-1} is added to the row $j = 0$ of (14), and the system is closed by

$$(u_1 - u_{-1}) + \frac{g}{3}\,(u_{-1} + 4u_0 + u_1) = 0, \qquad g = \gamma\Delta\eta. \quad (15)$$

This is (12) written with the compact (Padé) pair $(u_1 - u_{-1})/(2\Delta\eta) = \psi_0' + \frac{\Delta\eta^2}{6}\psi_0''' + \ldots$ and $(u_{-1} + 4u_0 + u_1)/6 = \psi_0 + \frac{\Delta\eta^2}{6}\psi_0'' + \ldots$ [41]. The $\Delta\eta^2$ terms combine into $\partial_\eta^2(\psi' + \gamma\psi)$. For frozen coefficients the equation turns ∂_η^2 into an operator along the ground, which annihilates (12), so the closure is fourth-order accurate. Relation (15) expresses u_{-1} through u_0 and u_1, so the system stays tridiagonal. The coefficients of row 0 at the ghost node are extrapolated quadratically, and the same closure is used in all $p + 4q$ factors of a step. The usual closure is the central difference $(u_1 - u_{-1})/(2\Delta\eta) + \gamma u_0 = 0$, as in the discrete mixed Fourier transform of split-step solvers [42]. We call it the mirror closure; it is second-order accurate.

VII. Discrete Dispersion Analysis

Freeze h_ξ, h_η and k, and consider a plane wave at angle θ to the local marching direction: $k_z = k\sin\theta$, computational

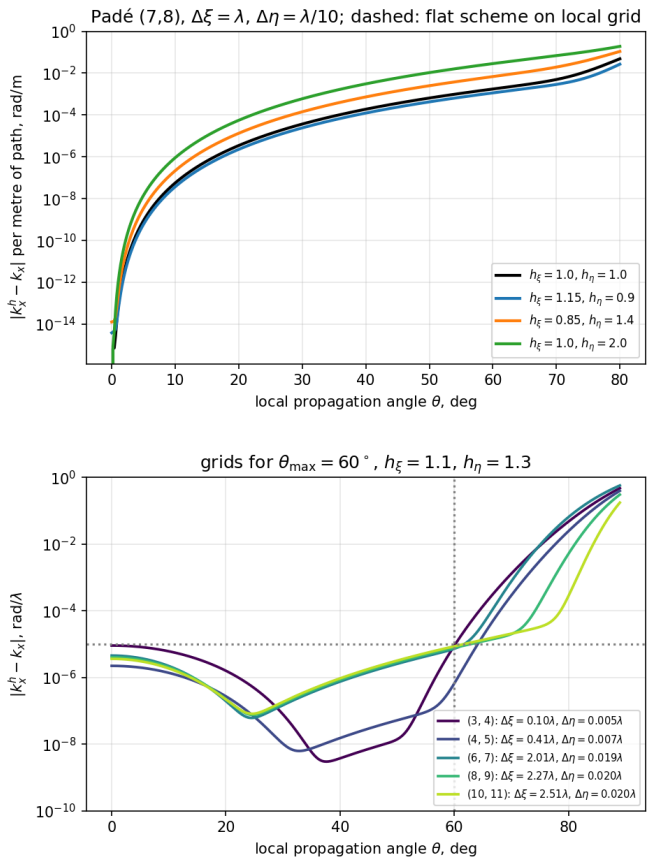


Fig. 2. Top: dispersion error for several metric values on one grid (solid) and of the flat scheme on the local effective grid (dashed); they agree to $4\cdot10^{-13}$ of the maximum error (Theorem 1). Bottom: Padé orders $(3,4)$–$(10,11)$ on grids selected for a 60° aperture in a distorted cell ($h_\xi = 1.1$, $h_\eta = 1.3$).

wavenumber $q = k_z h_\eta$, $\psi_j^n = \lambda^n e^{iqj\Delta\eta}$. The flux factors cancel. With $s = \sin^2(q\Delta\eta/2)$, the symbol of (14) is

$$L_h(q) = \nu - \frac{4}{\tau^2 r^2}\,\frac{s}{1-4\alpha s}, \qquad \lambda = R\left(L_h(q)\right), \tag{16}$$

and the exact factor is $e^{i\Delta\xi(p-\beta)}$ with $p = h_\xi k\cos\theta$. The dispersion error per unit physical length is

$$\varepsilon(\theta) = \frac{\left|\log\left(\lambda e^{-i\Delta\xi(p-\beta)}\right)\right|}{h_\xi\Delta\xi}. \tag{17}$$

Theorem 1 (Local equivalence): $\varepsilon(\theta;\beta,\Delta\xi,\Delta\eta,h_\xi,h_\eta) = \varepsilon_0(\theta;\beta/h_\xi,h_\xi\Delta\xi,h_\eta\Delta\eta)$, where ε_0 is the dispersion error of the flat scheme (4) with the given reference wavenumber and steps.

Proof: Let $\beta_e = \beta/h_\xi$, $\Delta x_e = h_\xi\Delta\xi$ and $\Delta z_e = h_\eta\Delta\eta$. Then:

- $\beta\Delta\xi = \beta_e\Delta x_e$, and the coefficients of R depend on this product only;
- $\nu = k^2/\beta_e^2 - 1$;
- $\tau r = \beta_e\Delta z_e$;
- $q\Delta\eta = k_z\Delta z_e$.

Hence L_h and λ coincide, and $e^{i\Delta\xi(h_\xi k\cos\theta-\beta)} = e^{i\Delta x_e(k\cos\theta-\beta_e)}$. ∎

Corollary 1 (Angles): The transformation introduces no angle restriction. A wave is resolved if its angle to the local marching direction lies within the aperture of the rational approximation on the local grid. Waves reflected by slopes of up to θ_s reach $\theta_{\max} + 2\theta_s$, so the aperture is set to that value. This requirement is physical and applies equally to a staircase.

Corollary 2 (Grid selection): Let $h_\pm = \max / \min h_\xi$. The flat-terrain optimizer [4], [5] is applied with:

- the wavenumber interval $[k_{\min}h_-, k_{\max}h_+]$;
- the vertical wavenumber bound $k_z^{\max}h_-$;
- the error budget per unit computational length.

The vertical step it returns is divided by $\max r\cdot h_+/h_-$.

Fig. 2 confirms Theorem 1 numerically. It also shows that every Padé order reaches the target accuracy up to the aperture in transformed coordinates.

A. Discrete Reflection by the Impedance Ground

The same frozen-coefficient analysis applies to the ground closure.

Proposition 2: Let a discrete wave $u_j = e^{-iqj} + R_h e^{iqj}$, $q = k\sin\phi\, h_\eta\Delta\eta$, satisfy the interior rows of (14). Then the closure (15) gives

$$R_h = \frac{iS(q)-g}{iS(q)+g}, \qquad S(q) = \frac{3\sin q}{2+\cos q} = q - \frac{q^5}{180} + O(q^7), \tag{18}$$

whereas the exact coefficient is $R = (iq-g)/(iq+g)$. The mirror closure gives (18) with $S = \sin q = q - q^3/6 + \dots$. If $\operatorname{Re}\Delta_g \geq 0$, then for both closures, with $\sigma = S(q)/q$,

$$|R_h - R| \leq \frac{2|1-\sigma|}{\sigma} \approx \begin{cases} q^4/90, & \text{compact,} \\ q^2/3, & \text{mirror,} \end{cases} \tag{19}$$

uniformly in the ground parameters.

Proof: Since the wave satisfies the interior rows, row 0 holds iff the ghost value $e^{iq} + R_h e^{-iq}$ satisfies (15), i.e. $2i\sin q\,(R_h - 1) + \frac{g}{3}(4+2\cos q)(1+R_h) = 0$, which is (18). Let $w = g/(iq) = \Delta_g/\sin\phi$. Then $R = (1-w)/(1+w)$, $R_h = (\sigma-w)/(\sigma+w)$ and $R_h - R = 2w(\sigma-1)/[(\sigma+w)(1+w)]$. For $\operatorname{Re} w \geq 0$ we have $|w| \leq |1+w|$ and $|\sigma+w| \geq \sigma$. ∎

The condition $\operatorname{Re}\Delta_g \geq 0$ holds for any passive ground with $\operatorname{Re}\varepsilon \geq 0$, for both polarizations. The bound depends only on the vertical wavenumber resolved by the grid, as the interior dispersion error does. A grid that resolves a wave in the interior therefore also reflects it accurately, whatever the ground. Fig. 3 shows $|R_h - R|$ for wet ground. At $k\Delta\eta = 0.5$ the compact closure is accurate to $6\cdot10^{-5}$ up to 60°, and the mirror closure to 10^{-2}.

VIII. Compatibility With the Nonlocal Boundary Condition

Theorem 2: Let $\eta_{\max} \geq \eta_\ell + 2\Delta\eta$, and let the medium above the transition layer be range-independent. Then the NLBC (5) computed for the chain R and the untransformed exterior scheme is the exact discrete transparent boundary condition of the scheme of Algorithm 1.

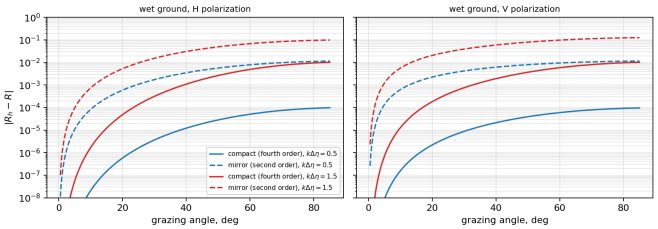


Fig. 3. Error of the discrete reflection coefficient of the impedance ground ($\varepsilon = 25 + 1.2i$, 300 MHz) for the compact closure (15) (solid) and the mirror closure (dashed), $h_\eta = 1$.

Proof: For $j\Delta\eta \geq \eta_\ell$ we have $h_\xi = h_\eta = r = m = 1$ and $V = 0$ exactly, because the taper is C^2. The rows there coincide with (4) with $z = \eta + z_{\mathrm{ref}}$. The flux pairs are the identity in this region, and are solved below it with the top value passed through. What reaches the boundary is therefore produced by the chain R alone, with exterior coefficients independent of n, and the derivation of [6] applies verbatim, including a linear (Airy) exterior profile. ∎

Remark 1: Placing the flux factors around R as in the second line of (13) is also correct, but the NLBC would then have to be computed for a chain of $p+2q$ factors containing reciprocal pairs. The matrix function of the inverse z-transform becomes defective, and the adaptive quadrature of the kernels does not converge. The bracketed form avoids this, and the kernels are the same as over flat ground.

IX. Comparison With Shift-Map Finite-Difference PEs

A. The Shift-Map Schemes as Rational Propagators

With $\zeta = z - T(x)$ and the gauge

$$u = v \exp\left(ik_0 T'\zeta + \tfrac{ik_0}{2}\int_0^x T'^2 dx'\right), \tag{20}$$

the NA PE over the terrain T becomes the NA PE over flat ground for v, with the index $n^2 - 2\zeta T''$ [8], [9]. The WA shift-map PEs apply the same transformation and replace the NA operator by a WA one [10], [16]. In the FD schemes surveyed in the introduction, the envelope satisfies $u_x = ik_0(f(X)-1)u$ with $X = k_0^{-2}\partial_\zeta^2 + n^2 - 1$, where f approximates $\sqrt{1+X}$ by one of:

- NA: $f = 1 + X/2$;
- Claerbout: $f = (1 + \frac{3}{4}X)/(1 + \frac{1}{4}X)$;
- Greene: $f = (0.99987 + 0.79624X)/(1 + 0.30102X)$;
- Padé $[2/2]$: $f = (1 + \frac{5}{4}X + \frac{5}{16}X^2)/(1 + \frac{3}{4}X + \frac{1}{16}X^2)$.

The range step is Crank–Nicolson,

$$u^{n+1} = \frac{1 + i\kappa(f(X)-1)}{1 - i\kappa(f(X)-1)}\,u^n, \qquad \kappa = \frac{k_0\Delta x}{2}. \tag{21}$$

X is discretized by three-point (second-order) or five-point (fourth-order) differences.

In this form the schemes are rational functions of the vertical operator, of degree $(1,1)$ or $(2,2)$, exactly like (3). The difference lies in what is approximated. Equation (21) first approximates the square root and then the exponential by the second-order Cayley transform, so its error grows with $k_0\Delta x$ and with the angle independently of the order of f. Equation (3) approximates the propagator $e^{i\beta\Delta x(\sqrt{1+L}-1)}$ itself, to order $2p$ in L. Its error vanishes uniformly as the order increases, for range steps of many wavelengths. The baseline solvers used below implement (20)–(21) in the same code as the proposed method, with the Numerov weight α set to zero. The two families therefore differ only in the approximation.

B. Grid-Independent Error of the Shift Map

For a constant slope $s = \tan\theta_s$, substituting a plane wave into the gauge (20) gives the error of the shift-map PE in the limit $\Delta x, \Delta\zeta \to 0$. A plane wave at the angle α to the horizontal has $v \propto e^{ik_0(\sin\alpha - s)\zeta}$. The PE propagates it with the longitudinal wavenumber $k_0[f(-(\sin\alpha - s)^2) - 1]$, whereas the exact wavenumber is $k_0(\cos\alpha + s\sin\alpha - 1 - s^2/2)$. Hence

$$\varepsilon_\infty(\alpha;\theta_s) = k_0\left|f\left(-(\sin\alpha - s)^2\right) + \frac{s^2}{2} - s\sin\alpha - \cos\alpha\right|, \tag{22}$$

after removal of the angle-independent part $f(0) - 1$. For $f = 1 + X/2$ this is $k_0|1 - \frac{1}{2}\sin^2\alpha - \cos\alpha|$, independent of the slope: the BT transformation is exact for the NA PE, and its floor is that of the NA PE on flat ground. For any other f, including the exact square root, the floor depends on the slope. It vanishes only on a narrow set of angles because the gauge is paraxial.

Fig. 4 shows (22) as a function of the angle to the slope. At an accuracy of 10^{-4} rad/λ (0.1 rad over $10^3\lambda$), the admissible angles are as follows.

- *Exact square root (any WA scheme):* only waves between 3.9° and 6.3° above the horizontal on a 10° slope, a single angle on a 20° slope, and none on a 30° slope.
- *Claerbout approximation:* the same windows on slopes. On these slopes the floor is set by the gauge, not by the square-root approximation, so raising the order of f cannot help.
- *NA PE:* $|\alpha| < 5.9°$ for every slope.

This quantifies the published observations: WA shift maps are limited to slopes of 10–20°, can be worse than the NA PE on lee-side slopes [24], [25], and their sheared vertical wavenumber is unphysical [15]. In the proposed coordinates, by Theorem 1, the frozen-coefficient error is that of the flat scheme on a local grid, and it vanishes under refinement for every slope.

C. Maximum Angle and Cost on Flat Ground

On flat ground the shift map plays no role, and the comparison concerns the propagators alone. For every method, accuracy budget $\varepsilon_{\max}$ and maximum angle $\theta_{\max}$, we searched by brute force over $10^{-2.5} \leq \Delta x/\lambda \leq 10^{2.5}$ and $10^{-3} \leq \Delta z/\lambda \leq 1$, and over $\beta \in [0.5, 1]k_0$ for the rational scheme. The cheapest grid is the one for which the dispersion error (17), with the same Numerov or FD symbols, stays below $\varepsilon_{\max}$ for all $\theta \leq \theta_{\max}$. The cost is the number of tridiagonal solves per λ^2, $p/(\Delta x\Delta z)$. The pentadiagonal systems of the five-point schemes are counted as one solve, which favors the FD methods. A phase drift common to all angles (Greene:

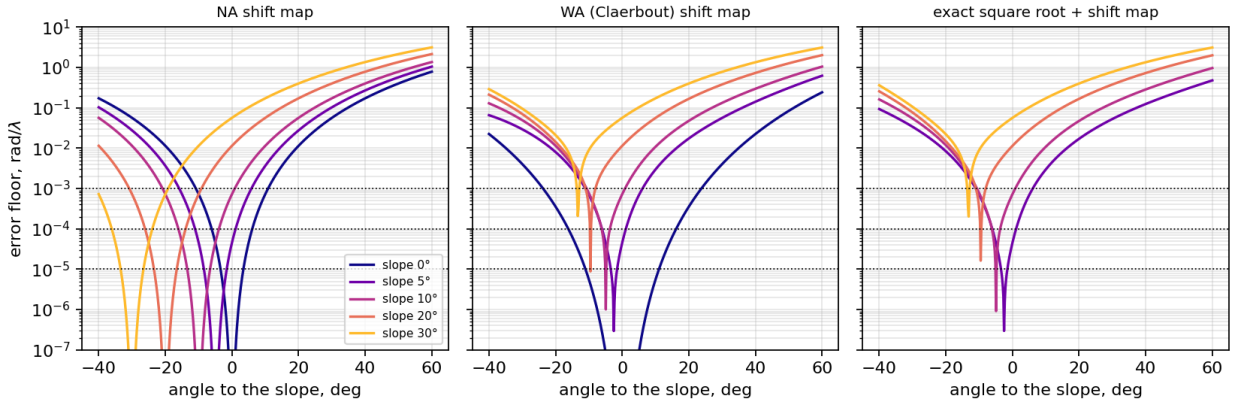


Fig. 4. Grid-independent error (22) of the shift map on a slope, versus the angle to the slope, for the NA approximation, the Claerbout approximation and the exact square root. Dotted: accuracy budgets of 10^{-3}, 10^{-4} and 10^{-5} rad/λ. For the proposed coordinates the corresponding error is zero.

TABLE I
MAXIMUM PROPAGATION ANGLE (DEGREES) MEETING AN ACCURACY BUDGET

budget, rad/λ (0.1 rad over)	10^{-3} ($10^2\lambda$)	10^{-4} ($10^3\lambda$)	10^{-5} ($10^4\lambda$)
NA-FD, CN, 3-point	10.8	6.0	3.3
WA-FD Claerbout, CN, 3-point	23.8	16.0	11.0
WA-FD Greene, CN, 5-point	11.8	3.3	1.0
WA-FD Padé $[2/2]$ sqrt, CN, 5-point	43.8	34.8	27.8
rational Padé $(1,2)$, Numerov	43.0	38.0	27.5
rational Padé $(2,3)$, Numerov	65.3	54.5	42.8
rational Padé $(4,5)$, Numerov	79.5	73.3	66.3
rational Padé $(6,7)$, Numerov	82.5	80.0	78.0
rational Padé $(8,9)$, Numerov	84.3	82.3	80.5

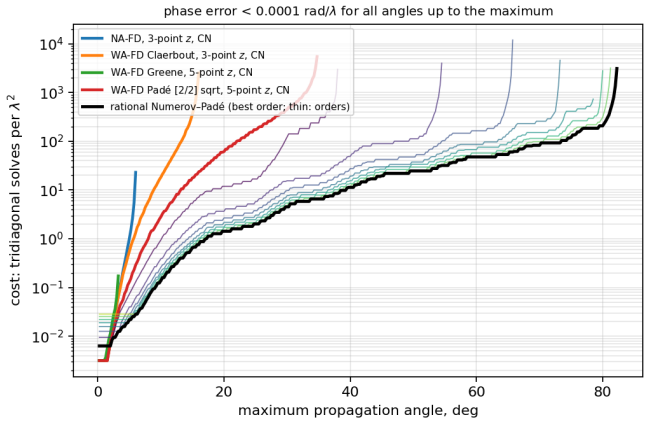


Fig. 5. Minimal cost (tridiagonal solves per λ^2) to keep the dispersion error below 10^{-4} rad/λ for all angles up to the maximum. Thick black: the rational scheme with the best order; thin lines: individual Padé orders $(1,2)$–$(8,9)$.

$f(0) \neq 1$; rational: $\beta \neq k_0$) is a global phase and is removed for all methods.

Table I and Figs. 5–6 show three things.

- *FD angle limits.* The FD PEs have an angle beyond which no grid meets the budget, because the error of f does not decrease with the grid (Fig. 6). At 10^{-4} rad/λ this angle is 6° (NA), 16° (Claerbout) and 35° (Padé $[2/2]$). The Greene approximation is a minimax fit with a relative error of about 10^{-4} over a wide range of angles; it is useful only for coarser budgets.
- *No angle limit for the rational scheme.* Its reachable angle grows with the order without bound, from 38° for $(1,2)$ to 82° for $(8,9)$, and reaches 80° even at 10^{-5} rad/λ. The search interval for Δz, not the method, ends the curves in Fig. 5.
- *Cost.* Where both are feasible, the rational scheme is far cheaper. At 10^{-4} rad/λ the cost ratio Claerbout/rational is 130 at 10° and 560 at 15°, and the ratio Padé-$[2/2]$-FD/rational is 20, 43 and 99 at 10°, 20° and 30°. The reason is mainly the range step. The Padé approximation of the propagator is accurate to high order in $\beta\Delta x$ and allows steps of the order of the wavelength or more. The Crank–Nicolson step is only second order in $k_0\Delta x(f-1)$, so its step must shrink quickly as the angle grows.

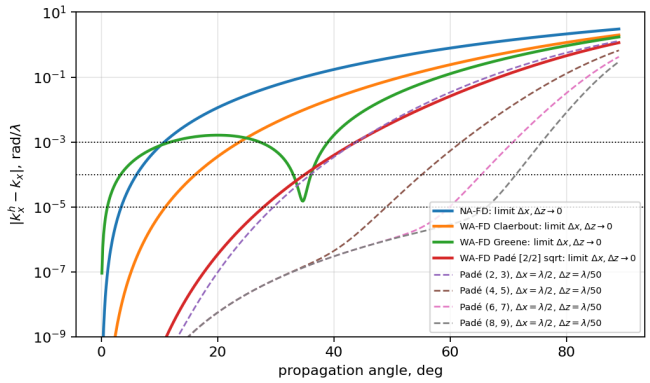


Fig. 6. Dispersion error versus angle: grid-independent limits of the FD PEs (solid) and the rational scheme on a coarse grid, $\Delta x = \lambda/2$, $\Delta z = \lambda/50$ (dashed).

D. Confirmation Against Exact Solutions

Beams over flat ground. Complex-source-point beams (300 MHz, Rayleigh parameter 50 m, divergence 2.3°) were launched at 5°–70° over a perfectly conducting ground and

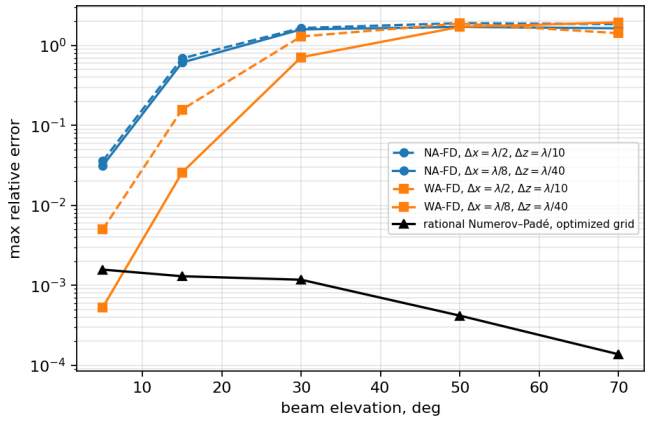


Fig. 7. Beams over flat perfectly conducting ground: maximum error with respect to the exact solution versus elevation, for NA-FD and WA-FD on two grids and for the rational scheme on its optimized grid.

propagated over 300 m. NA-FD and WA-FD (Claerbout) were run on a typical grid ($\lambda/2 \times \lambda/10$) and on a fine grid ($\lambda/8 \times \lambda/40$). The rational scheme used the grid chosen by its optimizer for the aperture $\theta+8°$ and precision 10^{-2}. All three use the NLBC on top, so only the propagator differs. Fig. 7 shows the maximum error with respect to the exact field.

- *FD PEs.* WA-FD is accurate at 5° but fails at 15° (2.6% on the fine grid) and beyond. NA-FD exceeds 3% already at 5°. Refining the grid does not help at 30° and above, in agreement with Table I.
- *Rational scheme.* Its error is $1.6 \cdot 10^{-3}$ to $1.4 \cdot 10^{-4}$ at all angles up to 70°. At 5° it uses 49 times fewer grid nodes than the typical FD grid, and 790 times fewer than the fine one.

Beam reflected by a slope. For the exact problem of Section XI-A the NA and WA shift maps were run on the fine grid ($\lambda/8 \times \lambda/40$). Their maximum errors are 0.67/0.56, 1.02/0.96, 1.41/1.68 and 1.19/1.29 (NA/WA) for slopes of 5°, 10°, 20° and 30°. The terrain-following rational solver gives $8.3 \cdot 10^{-5}$, $1.4 \cdot 10^{-4}$, $1.2 \cdot 10^{-3}$ and $2.9 \cdot 10^{-3}$. Already at 5° the reflected beam (20°) lies outside the admissible windows of Fig. 4.

X. Discrete NLBC Versus Absorbing Layers

Shift-map and split-step PEs truncate the domain by absorbing layers. The complex-index layer of [14] is $n^2 \to n^2 + iA\left((z - z_a)/D\right)^2$ over a layer of thickness D above the height of interest H, with $\psi = 0$ on top; window filters [27] act similarly. Such a layer works for a range of angles that depends on A and D:

- a weak or thin layer lets steep waves reach the top and reflect;
- a strong one reflects grazing waves at its gradient.

The discrete NLBC has no parameters: it is exact for the discrete scheme at every angle.

To isolate the boundary error we used the same propagator in all runs (Padé $(7, 8)$, Numerov, $\Delta x = 2.5\,\mathrm{m}$, $\Delta z = 0.1\,\mathrm{m}$ at 300 MHz), and only the top boundary was changed. On this grid the propagation error is small. With the NLBC, a domain 200 m taller gives the same field to all digits. The setup is:

- beams at 1°, 3°, 10° and 30° leave the region of interest $z < H = 200\,\mathrm{m}$ through the top;
- the ranges are 17.2, 5.7, 1.7 and 0.5 km, so that each beam crosses H well inside the domain;
- the NLBC is placed at $H + 20\,\mathrm{m}$;
- the absorbing layers have $D = H/4$, $H/2$ and H, and $A = 10^{-3}$–1.

Fig. 8 shows the maximum error in $z < H$.

- *NLBC accuracy.* With the NLBC the error is $4 \cdot 10^{-7}$, $1.4 \cdot 10^{-6}$ and $4 \cdot 10^{-5}$ for 1°, 3° and 10°. At 30° it is $4.6 \cdot 10^{-3}$, the propagation error of this grid at that angle, which every boundary shares.
- *Absorbing layers.* Their errors depend strongly on the angle and on (A, D). For grazing beams no setting comes close to the NLBC. At 1° the best layer ($D = H$, $A = 10^{-3}$) gives $1.6 \cdot 10^{-3}$, four thousand times the NLBC error, and stronger layers give up to $8 \cdot 10^{-2}$. At 3° the best value is $4.7 \cdot 10^{-5}$ against $1.4 \cdot 10^{-6}$.
- *Steep beams.* Weak or thin layers reflect, with errors up to 0.28 for $D = H/4$.
- *No universal setting.* The value $A = 10^{-2}$, $D = H$ is acceptable from 3° upward, but gives $7.5 \cdot 10^{-3}$ at 1°. The thinner layers recommended in the literature ($D \approx 0.3H$) are worse.

The absorbing layer therefore needs tuning to the angular content of the field, which is not known in advance. The NLBC needs none.

Cost. A layer of thickness $D = H$ doubles the number of vertical nodes; here, 4001 instead of 2201. The NLBC adds a one-time computation of the convolution kernels, and at each step a convolution with the history of the boundary values, $O(p^2 N_x)$ operations, independent of the number of vertical nodes. With the small vertical grids of this test these overheads dominate. The NLBC runs took 71, 11, 4.3 and 2.8 s for 1°, 3°, 10° and 30° ($N_x = 6876$, 2291, 682 and 209). The $D = H$ layers took 18–22, 5, 1.3 and 0.4 s. The relative overhead decreases with the number of vertical nodes. Recursive or sum-of-exponentials approximations of the convolution [23], [43] remove its growth with N_x. The gain from the NLBC is therefore reliability and accuracy, not speed on small grids.

XI. Numerical Results

The method is implemented in JAX in the open-source library `pywaveprop` [44], as a subclass of its rational propagator. It reuses the Padé coefficients, the grid optimizer and the NLBC. Over flat ground it reproduces the existing propagator, including the NLBC, to $4 \cdot 10^{-14}$.

A. Exact Solution: Beam Reflected by a Conducting Slope

The incident field is a 300 MHz complex-source-point beam, an exact solution of the Helmholtz equation. Its waist is at $(-50, 30)$ m, its Rayleigh parameter is 100 m, and it is directed at −10°. Its image in the line $z = x \tan\theta_s$ gives the exact field above the Dirichlet slope; the reflected beam leaves

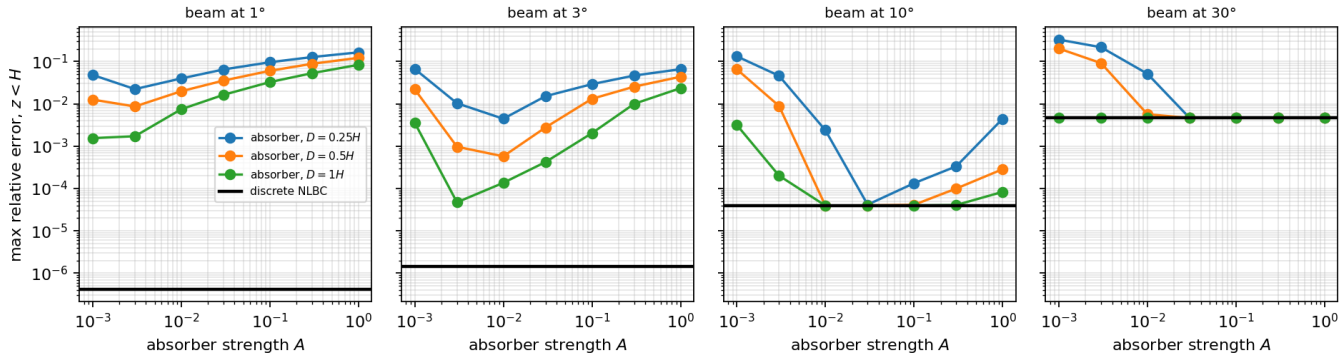


Fig. 8. Maximum error below H for beams leaving the domain through the top: absorbing layers of thickness D and strength A versus the discrete NLBC (black). Same propagator and grid in all runs.

TABLE II
EXACT SLOPE TEST: RELATIVE MAXIMUM ERROR

		terrain-following			staircase		
θ_s	beam	Δx, m	Δz, m	error	Δx, m	Δz, m	error
10°	30°	2.5	0.035	$1.4 \cdot 10^{-4}$	2.5	0.063	0.54
20°	50°	2.5	0.019	$1.2 \cdot 10^{-3}$	2.5	0.035	0.69
30°	70°	1.0	0.009	$2.9 \cdot 10^{-3}$	1.0	0.016	0.81

TABLE III
EXACT IMPEDANCE-SLOPE TEST: RELATIVE MAXIMUM ERROR

θ_s	ground	Δx, m	$\Delta\eta$, m	H	V
5°	wet	2.5	0.063	$4.4 \cdot 10^{-5}$	$3.9 \cdot 10^{-5}$
10°	wet	2.5	0.037	$9.2 \cdot 10^{-5}$	$3.0 \cdot 10^{-5}$
10°	dry	2.5	0.037	$8.2 \cdot 10^{-5}$	$3.8 \cdot 10^{-5}$
10°	sea	2.5	0.037	$9.6 \cdot 10^{-5}$	$8.1 \cdot 10^{-5}$
20°	wet	2.5	0.020	$8.1 \cdot 10^{-4}$	$8.2 \cdot 10^{-4}$
30°	wet	1.25	0.010	$9.3 \cdot 10^{-4}$	$9.2 \cdot 10^{-4}$

at $2\theta_s + 10°$. Both solvers start from the exact field, use Padé $(7, 8)$ and the aperture $\theta_{\max} = 10° + 2\theta_s + 5°$, and their grids are chosen by the respective optimizers with precision 10^{-2}. Table II gives the maximum error relative to the maximum of the exact field, and Fig. 9 shows the fields and the errors.

The terrain-following error is 10^{-4}–$3{\cdot}10^{-3}$ up to a reflected beam at 70°. The staircase loses most of the reflected beam (Fig. 9), and its error decreases only slowly under refinement. For $\theta_s = 20°$ it is 0.69, 0.55 and 0.54 at $\Delta z = 0.035$, 0.013 and 0.005 m ($\lambda/200$), and still 0.45 when in addition Δx is reduced to 0.625 m. The terrain-following grids have 1.2–1.4 times more vertical nodes, because of the expansion $\max r$ in the transition layer. Each step also costs $p + 4q$ instead of p solves, so the run time was 7–11 times that of the staircase, at errors two to three orders of magnitude smaller.

B. Role of the Flux-Conserving Step

For $\theta_s = 20°$, three variants of the range step give:

- Liouville rescaling alone ($q = 0$): error $6.4 \cdot 10^{-2}$, an amplitude drift along the reflected beam;
- unrotated $[2/2]$ flux factors: a spurious evanescent component grows to 0.14;
- rotated flux factors: $1.2 \cdot 10^{-3}$.

Both ingredients of Section V are therefore necessary.

C. Exact Solution: Beam Reflected by an Impedance Slope

For the impedance ground the incident beam is written as a superposition of plane waves with the Gaussian angular spectrum $\exp[-((\phi - \phi_0)/2°)^2]$, focused at $(-50, 30)$ m and directed at $-5°$ (waist about 9 m). Each plane wave is reflected by the slope with the coefficient $R(\phi)$ of (2), so the sum is an exact solution of the Helmholtz equation with the impedance condition on the slope. The spectrum is truncated where it falls below e^{-36}; the truncated sum is still exact. We take three grounds at 300 MHz: wet ground ($\varepsilon = 25 + 1.2i$), dry ground ($4 + 0.006i$) and sea water ($70 + 300i$). For V polarization over wet ground the Brewster angle is 11°, and the reflected beam is weak and sensitive to the impedance. The solver uses Padé $(7, 8)$, the aperture $5° + 2\theta_s + 8°$, and the grid of its optimizer for precision 10^{-2}. Table III gives the error over the whole domain.

The error is below 10^{-3} for all grounds and both polarizations, up to a reflected beam at 65°, and is of the same size as for the conducting slope. To isolate the ground closure, the case $\theta_s = 10°$ over wet ground was repeated on prescribed grids ($\Delta x = 2.5$ m, $\beta = k \cos 20°$) with both closures:

$k\Delta\eta$	0.39	0.63	1.05	1.57
V, compact	$7.5 \cdot 10^{-5}$	$2.5 \cdot 10^{-4}$	$1.7 \cdot 10^{-3}$	$8.6 \cdot 10^{-3}$
V, mirror	$7.6 \cdot 10^{-4}$	$1.9 \cdot 10^{-3}$	$5.4 \cdot 10^{-3}$	$1.3 \cdot 10^{-2}$
H, both	$2.6 \cdot 10^{-4}$	$8.1 \cdot 10^{-4}$	$5.4 \cdot 10^{-3}$	$2.7 \cdot 10^{-2}$

For V polarization the mirror closure dominates the error on fine grids, and the compact closure reduces it by a factor of 3–10. For H polarization $|w| = |\Delta_g| / \sin\phi \approx 20$, so $|R_h - R| \approx 2|1 - \sigma|/|w|$ is small for both closures, and the error is that of the interior scheme. The compact closure costs nothing extra, since the system remains tridiagonal.

D. Real Terrain and NWP Refractivity

A 300 MHz horizontally polarized antenna on a 30 m mast stands on top of a ridge in the Scottish Highlands (57.2163°N

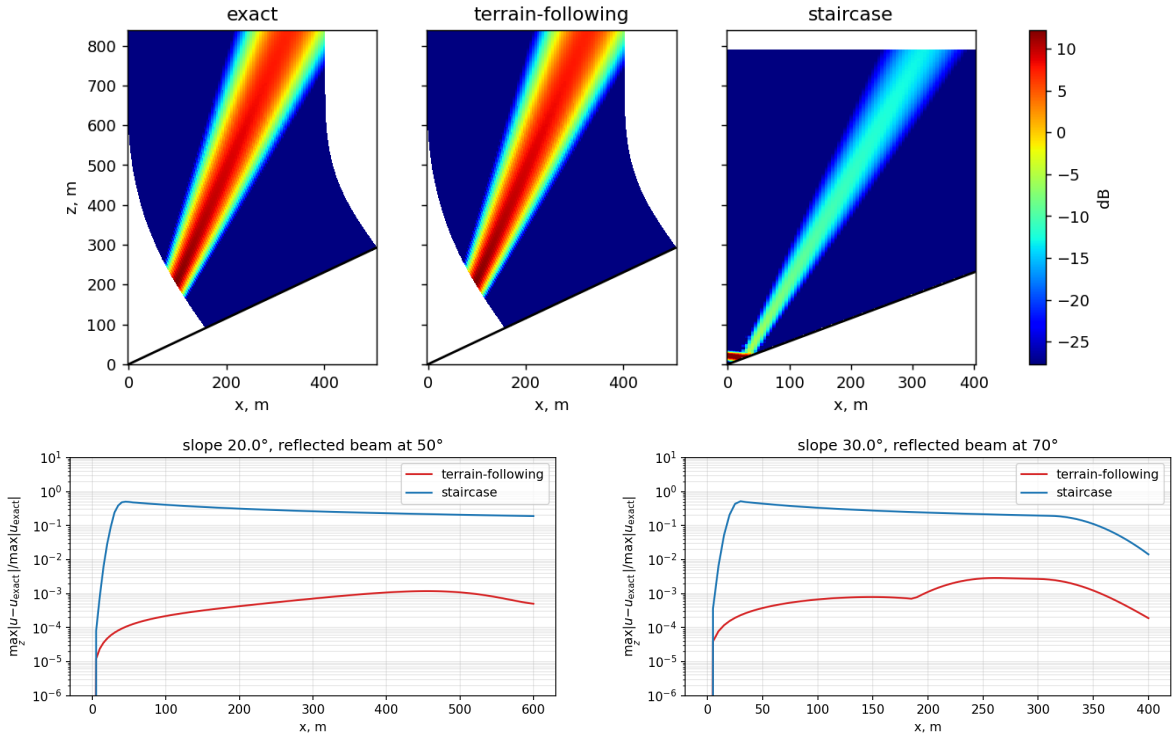


Fig. 9. Beam reflected by a perfectly conducting slope. Top: $|u|$ for $\theta_s = 30°$ (reflected beam at 70°); the terrain-following domain is bounded by the curved trajectories $\xi = 0$ and $\xi = x_{\max}$. Bottom: maximum error over height versus range for $\theta_s = 20°$ and 30°.

4.9214°W) and radiates along a 28 km path to 57.40°N 4.60°W. Its beam is 4° wide and tilted 1° down.

- *Terrain:* SRTM-type elevation at 0.0005°, from 166 to 664 m, smoothed with $\sigma = 100$ m; the maximum slope is 11.1°.
- *Refractivity:* the modified refractivity of the GFS analysis of 2026-10-09 12 UTC, interpolated along the path and averaged over range.
- *Ground:* perfectly conducting.
- *Grid parameters:* aperture $5° + 2 \cdot 11.1°$, transition height $L = 1000$ m, $z_{\text{ref}} = \overline{B}$.

Both solvers see the same smoothed terrain. Fig. 10 shows the path loss.

Each solver is converged with respect to its own grid. The two precision settings (10^{-2} and 10^{-3}) differ, at 10 m above the ground, by 0.08 dB (median) for the terrain-following solver and by 0.15 dB for the staircase. The two solvers differ by 1.1 dB at 10 m and 0.44 dB at 50 m above the ground. The largest differences, up to a few decibels, occur on the slopes facing the transmitter, where the staircase solution carries small-scale ripple consistent with scattering by the steps (Section XI-A). The terrain-following runs took 468 and 1390 s on a CPU, against 73 and 117 s for the staircase. The terrain-following grid ($\Delta x = 3.1$ m, $\Delta\eta = 0.034$ and 0.019 m) is finer because of the metric spread $h_\xi \in [0.91, 1.16]$ and the expansion $\max r = 1.28$, which enter the grid selection through the global bounds of Corollary 2.

XII. Discussion and Limitations

- *Cost.* The grid is chosen with the extreme values of h_ξ and r over the whole domain, which is pessimistic for strongly distorted maps; a local error budget or a map minimizing the spread of h_ξ would reduce it.
- *Smoothness.* The terrain must be C^2. Knife edges, cliffs and overhangs are smoothed over the scale σ, and very steep relief requires a larger transition height.
- *Ground impedance.* The Leontovich condition is local and angle independent. It is accurate for $|\varepsilon| \gg 1$, but not for dry ground at steep incidence. An exact angle-dependent lower NLBC [7] over terrain would require a kernel for range-dependent boundary coefficients, which is an open problem.
- *Refractivity.* The medium above the transition layer must not depend on range, as for any discrete NLBC. A range-dependent medium inside the layer is admissible, since the coefficient tables are per column.

XIII. Conclusion

Orthogonal terrain-following coordinates, built from the tapered harmonic extension of the terrain and its orthogonal trajectories, turn irregular terrain into a variable coefficient and a weight in the vertical Numerov scheme. Rational approximations of arbitrary order then apply with the angle limit of the flat scheme on a local effective grid, and the discrete NLBC is used without modification. A flux-conserving step with rotated-branch-cut factors removes the amplitude drift caused by the rotation of the coordinate lines. On an exact

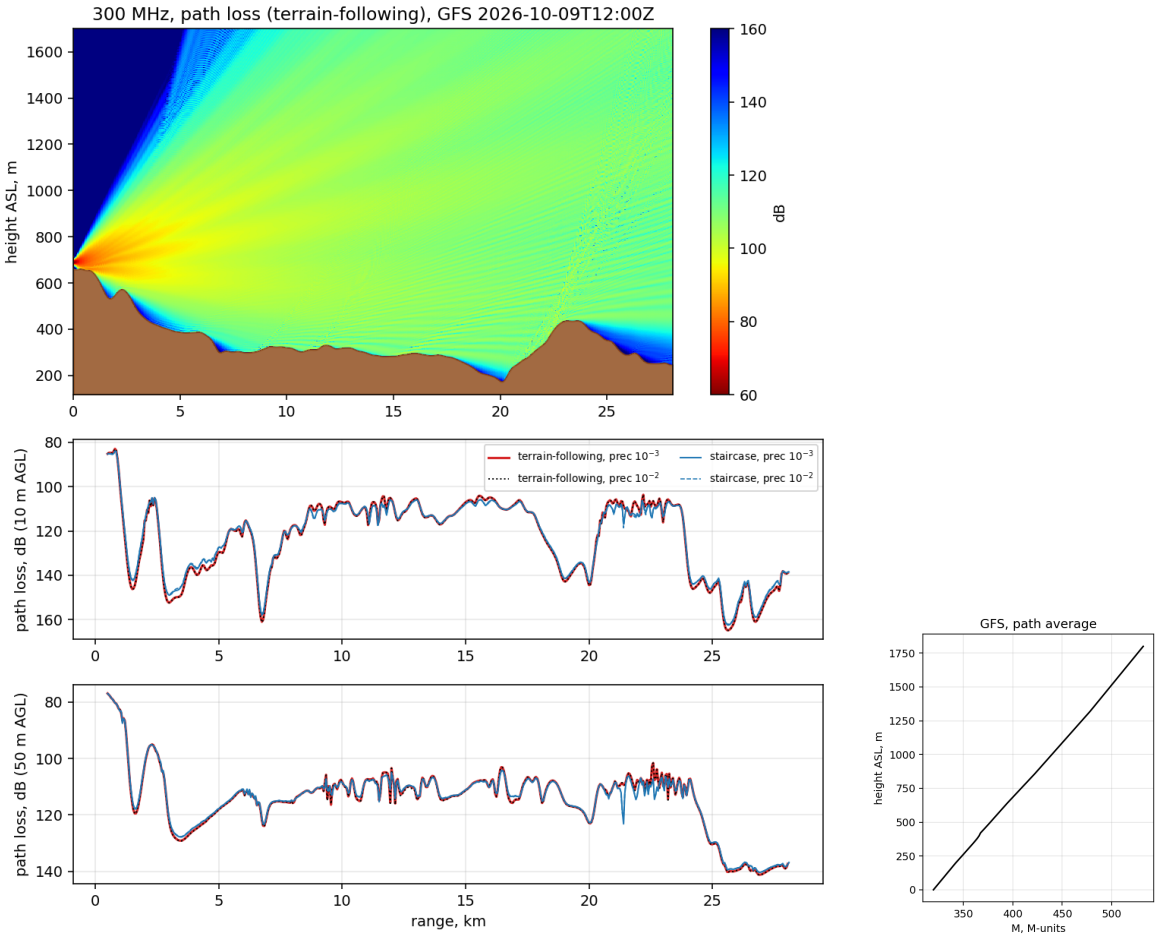


Fig. 10. Scottish Highlands, 300 MHz. Left: path loss of the terrain-following solver, and path loss 10 m and 50 m above the ground for both solvers at two precision settings. Right: GFS modified refractivity averaged along the path.

wide-angle benchmark the method is two to three orders of magnitude more accurate than the staircase. A compact closure of the Leontovich condition reflects every resolved wave with an error bounded uniformly in the ground parameters, and keeps the error below 10^{-3} over impedance slopes for both polarizations. Compared with the shift-map finite-difference PEs, it has no grid-independent slope error and no maximum propagation angle, and it costs one to three orders of magnitude less at moderate angles. Unlike absorbing layers, the discrete transparent boundary needs no tuning and is accurate for grazing as well as steep waves.

Code Availability

The implementation, tests and the scripts reproducing all figures are available in the `pywaveprop` repository [44].

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